

Lecture 1: Basic Concepts in Time Series Analysis

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Ghent University

Organizational Issues

- **Instructor:** Ko Sugiura.
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 - **Office hours:** by appointment only
- **Course website:** UFORA (F000955A - Applied Econometrics II)
- **Course References:** Mostly slides, but I will include additional references in case there's other material you can review.

Organizational Issues

- **Time and location:**
 - Thursday 11:00-12:15, Room 0.4, Blandijn, Campus Boekentoren
 - Friday 16:00-17:15, Auditorium A - Jean-Norbert Cloquet, T2, Campus Ufo
- **Course Schedule (tentative):** 10 lectures + 1 review session.

	Topic	Dates
1.	Basic Concepts in Time Series Analysis	12/02, 20/02, 27/02
2.	Linear Regression: Finite Sample Properties	06/03
3.	Linear Regression: Large Sample Properties	13/03, 19/03
4.	Unit Root, Cointegration	27/03
5.	Forecasting	03/04
6.	VAR Modelling	23/04
7.	Local Projections	30/04
	Final review class	08/05

* Class room will be different on 03/04.

Organizational Issues

- **Grading:**

- **Problem sets (30%):** Posted on UFORA. Involve both paper-and-pencil and computer exercises. You may use the programming language you choose (suggested: R, STATA, Python).

	Topics	Posted	Due Date
Assignment 1	1.	27/02	13/03
Assignment 2	2., 3., and 4.	27/03	10/04
Assignment 3	5., 6., and 7.	30/04	15/05

- **Final Exam (70%): June 12, 2026.**

Basic Concepts in Time Series Analysis

Introduction

What is a time series?

Definition (time series)

A **time series** is a collection of observations $X_1, X_2, X_3, \dots, X_T$, with each one being recorded at a specific time t .

- Informal definition: Collection of random variables indexed by time.
- Focus on a discrete time interval: the time elapsed between two observations is assumed to be constant (e.g., daily data, weekly data, annual data).
- Notation: $\{X_t\}_{t=1}^T$.

Objectives in time series analysis

- Time series data are useful for
 - Description
 - Explanation
 - Hypothesis testing
 - Simulation
 - Forecasting

Objectives in time series analysis

- Time series data are useful for
 - **Description:**
 - * Visualization of the data
 - * Eliminate “obvious” components in the data
 - * Leave “unobvious” components in the data
- ⇒ **Main focus of today's class**

- Explanation
- Hypothesis testing
- Simulation
- Forecasting

Objectives in time series analysis

- Use time series data for
 - Description
 - **Explanation:**
 - * Use the past data to understand the time-series pattern.
 - * Establish a time-series model.
 - * May be implied by the underlying theory (e.g., economic theory).
 - Hypothesis testing
 - Simulation
 - Forecasting

Objectives in time series analysis

- Use time series data for
 - Description
 - Explanation
 - **Hypothesis testing:**
 - * “Judge” the statement about the mechanism between the variables under study.
 - Simulation
 - Forecasting

Objectives in time series analysis

- Use time series data for
 - Description
 - Explanation
 - Hypothesis testing
 - **Simulation:**
 - * Use the past data to calculate past/present values of the variables in a hypothetical environment.
 - * Study counterfactual scenarios.
 - * e.g., What would have happened to GDP if tariffs were raised by 50%?
 - Forecasting

Objectives in time series analysis

- Use time series data for
 - Description
 - Explanation
 - Hypothesis testing
 - Simulation
 - **Forecasting:**
 - * Use the past data to calculate future values of the variables.
 - * e.g., What will GDP look like in 2035?

Basic Concepts in Time Series Analysis

Description: Visualizing our Data

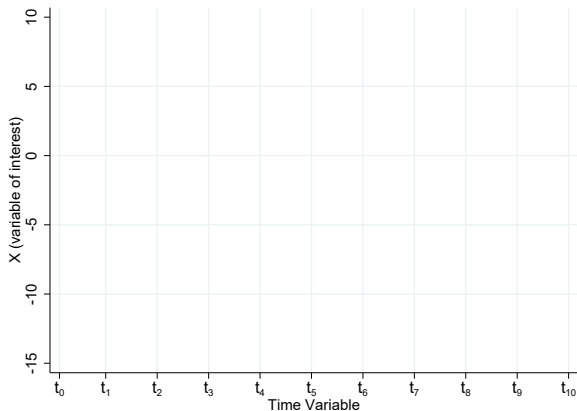
Time Series Visual Representation

- Most important visualization tool: **Time series plot**

Should be the first step analyzing times series $\implies$ **ALWAYS PLOT YOUR DATA!!**

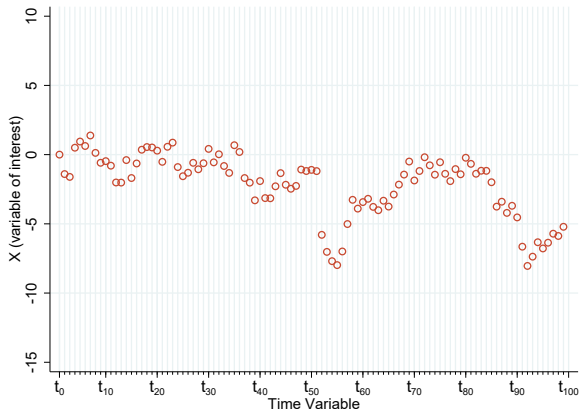
Time Series Visual Representation

- Most important visualization tool: **Time series plot**
 - **Horizontal axis:** time (an index set taking integer values)
 - **Vertical axis:** variable of interest (positive/negative continuous or discrete values)



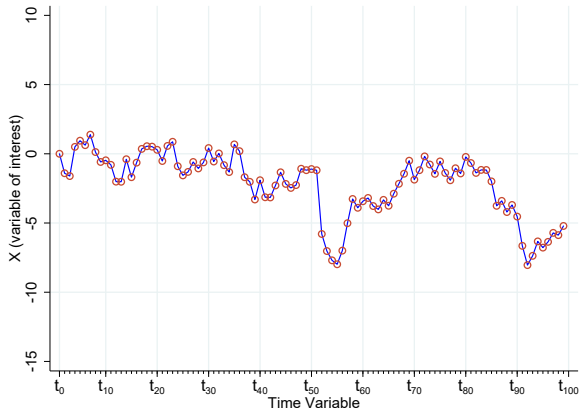
Time Series Visual Representation

- Most important visualization tool: **Time series plot**
- Each dot represents a single observation.



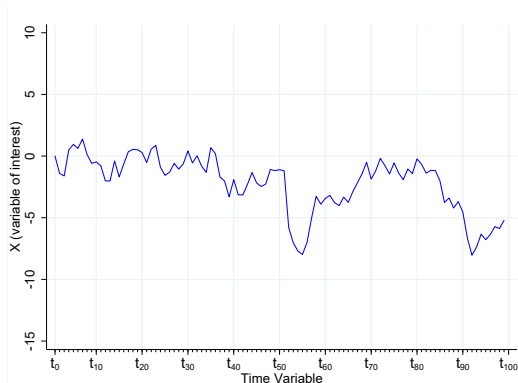
Time Series Visual Representation

- Most important visualization tool: **Time series plot**



Time Series Visual Representation

- When doing a **Time series plot**
 - **Visualization:** Usually (but not always) keep the continuous line
 - **Scale, intercept, range of the axes:** can change message and be deceptive
 - **Descriptive information:** titles, units of measure, time frequency, legend, etc



Basic Concepts in Time Series Analysis

Three Examples

Example 1: S&P500

STATA code

R code

Standard and Poor's 500: Stock price index from 500 largest U.S. companies

Time variable: Daily frequency.

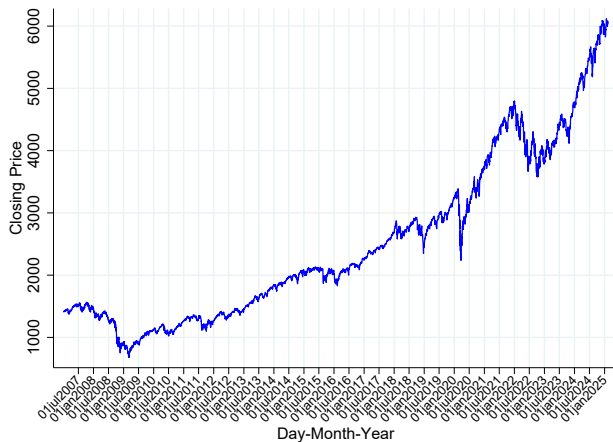


Figure 1: S&P500 Closing price

Example 1: S&P500

[STATA code](#)[R code](#)

- **S&P 500** exhibits an increasing pattern over time $\implies$ **time trend**

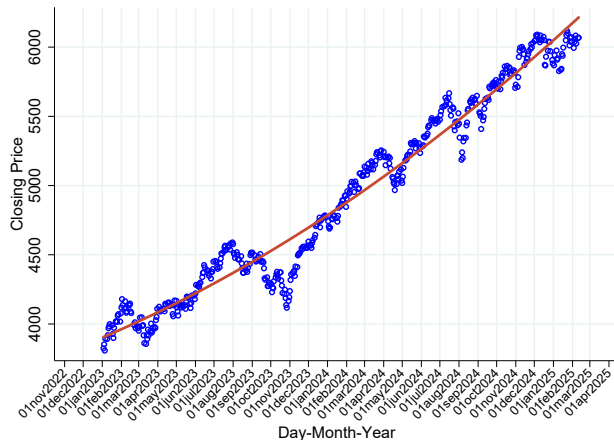


Figure 1: S&P500 Closing price

Example 1: S&P500

STATA code

R code

- **S&P500** shows a lot of day-to-day **fluctuations**
- But **dependence across time**: high/low price today $\Rightarrow$ high/low price tomorrow

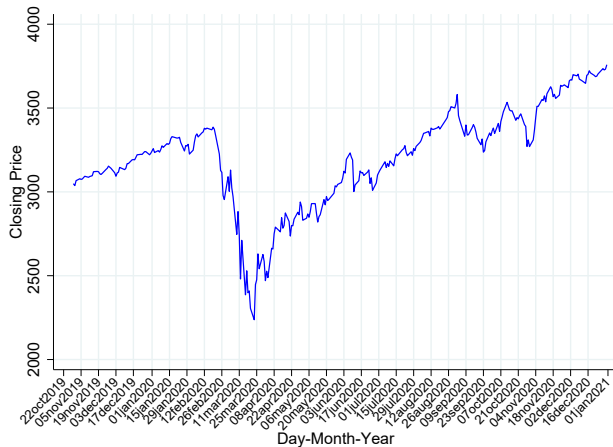


Figure 1: S&P500 Closing price

Example 2: Energy Production

- **Energy Production:** Industry-weighted index of electric power production for the U.S.
 - **Time:** Quarterly frequency.

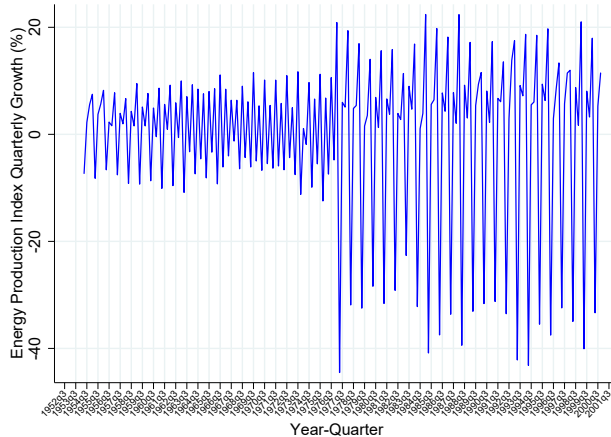


Figure 2: Energy Production Quarterly Growth

Example 2: Energy Production

- A flat-looking series, without trend
- **Striking change in volatility** when comparing pre and post 1976 periods

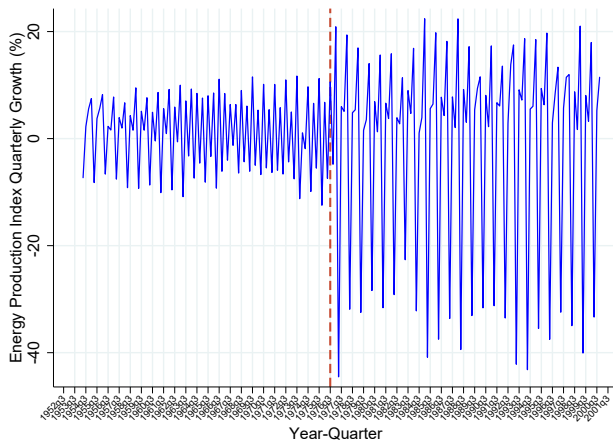


Figure 2: Energy Production Quarterly Growth

Example 2: Energy Production

- Another particularity $\Rightarrow$ **a regular pattern between quarters**

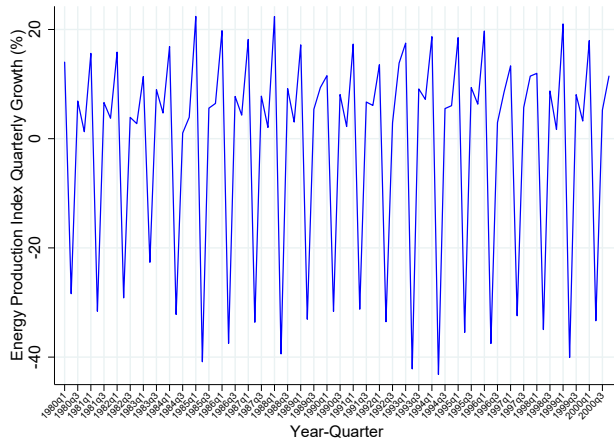
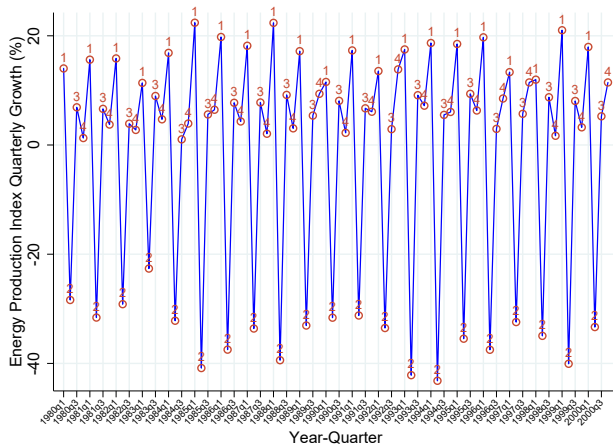


Figure 2: Energy Production Quarterly Growth

Example 2: Energy Production

- Another feature $\Rightarrow$ **a regular pattern between quarters**
 - Decreases in Q2 (i.e., Apr-Jun)
 - Increases in Q1, Q3, and Q4 (i.e., Jan-Mar, Jul-Sep, Oct-Dec)



Example 3: Energy and Consumption Goods

- Comparing energy and consumption goods production

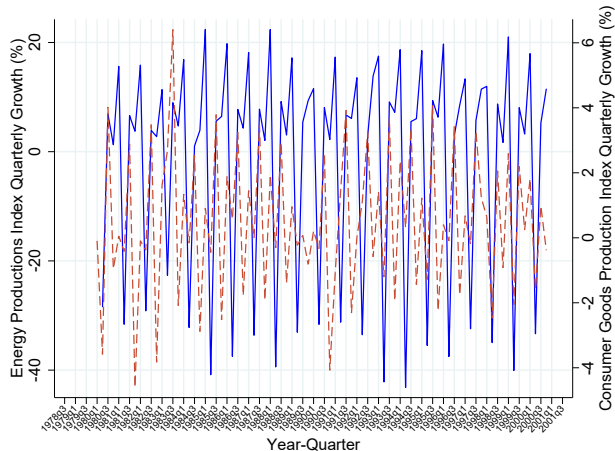


Figure 3: Energy and Cons. Goods Production Quarterly Growth

Example 3: Energy and Consumption Goods

- These variables are positively **correlated**

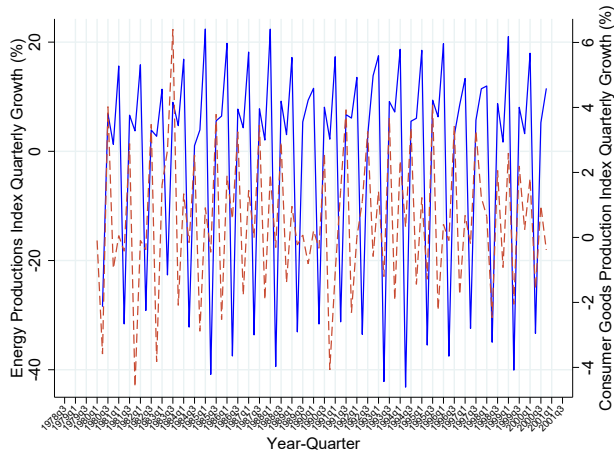


Figure 3: Energy and Cons. Goods Production Quarterly Growth

Main components of a time series

- All examples point to several characteristics of time series data
 - Trends (apparent in S&P500)
 - Seasonal or cyclical effects (apparent in Energy Production Growth)
 - Dependence and random fluctuations (all Examples)

Main components of a time series

- All examples point to several characteristics of time series data
 - Trends (apparent in S&P500)
 - Seasonal or cyclical effects (apparent in Energy Production Growth)
 - Dependence and random fluctuations (all Examples)
- Classical approach in time series analysis postulates that $\{X_t\}_{t=1}^T$ can be divided into:
 - Trend component (m_t)
 - Seasonal component (s_t)
 - Stochastic process ($\tilde{X}_t$)

Basic Concepts in Time Series Analysis

Trend, Seasonal, and Stochastic Components

About Trend and Seasonal Components

- Classical decomposition:

$$X_t = \underbrace{m_t}_{\text{Trend component}} + \underbrace{s_t}_{\text{Seasonal component}} + \underbrace{\tilde{X}_t}_{\text{Stochastic process}}$$

- Decomposition is **not unique in general**.
- We will need to make **assumptions** about the **structure**.

About Trend and Seasonal Components

- We will focus on the **stochastic component**.
- Why do we assume away the **trend** and **seasonal** components?
 1. These two components can be easily detected using a time-series plot.
 - These are less likely be objects of interest.
 2. We want to study the “well-behaved” probabilistic/statistical pattern of $\{X_t\}_{t=1}^T$.
 - Our statistical analysis hinges on the “well-behaved” assumption.
 - Adding the **trend** and **seasonal** components makes $\{X_t\}_{t=1}^T$ a “non-well-behaved” process.
- Our focus: $X_t = \underbrace{\tilde{X}_t}_{\text{Stochastic process}}$

Summary and Preview

Summary

- Time series $\{X_t\}_{t=1}^T$ is a collection of observations indexed by time
 - Time is discrete and equally divided.
- Classical decomposition:

$$X_t = \underbrace{m_t}_{\text{Trend component}} + \underbrace{s_t}_{\text{Seasonal component}} + \underbrace{\tilde{X}_t}_{\text{Stochastic process}}$$

- We will focus on a special case:

$$X_t = \underbrace{\tilde{X}_t}_{\text{Stochastic process}}$$

Preview

- We will clarify the meaning of “well-behaved” probabilistic/statistical pattern of $\{X_t\}_{t=1}^T$.
 - **Strong stationarity** and **weak stationarity**
- Three key aspects of a probability distribution:
 - **Mean, variance, and covariance**